2026 年（第十三届）英特尔杯大学生电子设计竞赛嵌入式 AI 专题赛

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作品题目	智驭低空枢纽：面向低空巡检与火源识别的多模态无人机地面站系统 Low-Altitude Intelligent Hub: A Multimodal UAV Ground Station System for Low-Altitude Patrol and Fire Detection		
作品简介	"智驭低空枢纽"面向低空巡检与火源识别场景，设计并实现了一套多模态无人机地面站系统。系统以 Intel DK-2500 地面端为核心，接入实时图传，结合镜头校正、火源检测、连续帧确认、任务地图、事件日志和 OpenVINO Whisper 语音识别，构建"图传感知—目标识别—候选指令—安全确认—飞控执行—状态回传—事件记录"的完整闭环。作品将语音、手势、视线和面板按钮统一转化为候选任务，并对起飞、巡线、返航、降落等高风险动作设置二次确认，降低误触发风险。系统采用低成本模块化飞行端与地面智能枢纽组合，将复杂飞控操作封装为任务级流程，降低操作门槛和培训成本。测试表明，系统可稳定完成图传接入、火源识别、多模态交互、任务地图联动和安全可追溯记录，适用于教学竞赛、原型验证和轻量级低空巡检场景。 Low-Altitude Intelligent Hub is a multimodal UAV ground station system designed for low-altitude patrol and fire detection. Built around an Intel DK-2500 ground terminal, the system integrates AMB82 real-time video streaming, lens correction, fire-source detection, multi-frame confirmation, mission-map visualization, event logging, and OpenVINO Whisper speech recognition. It forms a complete closed loop from video perception, target detection, candidate command generation, safety confirmation, flight-control execution, telemetry feedback, to event recording. Voice, gesture, gaze, and panel-button inputs are unified into candidate tasks, while high-risk actions such as takeoff, patrol, return-to-home, and landing require secondary confirmation to reduce accidental triggering. By combining a low-cost modular flight platform with an intelligent ground hub, the system encapsulates complex flight-control operations into task-level workflows, lowering training and operation costs. Tests show that the system can stably support video access, fire detection, multimodal interaction, mission-map linkage, and traceable safety records, making it suitable for teaching competitions, prototype validation, and lightweight low-altitude patrol applications.		